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Academic course planning recommendation and students' performance prediction multi-modal based on educational data mining techniques

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Abstract

Educational Data Mining (EDM) has recently received significant attention, leading to the development of various Data Mining (DM) methodologies for extracting hidden knowledge within educational data. This knowledge is crucial for enhancing teaching methods and improving student learning experiences, ultimately contributing to better student performance and overall educational outcomes. Students confront difficulties in selecting appropriate courses and suitable departments, which is regarded as the most important factor in avoiding career failure. Predicting students' academic performance is vital for evaluating the success of educational institutions. In this study, eleven Machine Learning (ML) algorithms and three Deep Learning (DL) algorithms namely Support Vector Classification (SVC), K-Nearest Neighbor (KNN), Logistic regression (LR), Decision tree (DT), Linear discriminant analysis (LDA), Quadratic Discriminant Analysis (QDA), Random Forest (RF), Gradient Boosting (GB), Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (Light GBM), Extra Trees, Deep Artificial Neural Network (DANN), Gated Recurrent Unit (GRU), and Long Short-Term Memory (LSTM), were evaluated using real dataset from the Faculty of Computers and Information Sciences (FCIS) at Mansoura University (MU). A prediction model was developed to predict students' academic grades in upcoming courses based on their past performance, alongside a recommendation model for guiding students towards suitable courses and departments. The results demonstrate that the Support Vector Classification (SVC) model outperformed others, achieving a 78.04% multi-classification accuracy and a 75.37% F1-Score. This study underscores the potential of individual ML and DL models to predict students' academic performance based on real dataset features.

Keywords Educational data mining (EDM) · Machine learning (ML) · Deep learning (DL) · Students' academic performance · Prediction model · Recommendation model

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Abbreviations

DM	Data Mining
EDM	Educational Data Mining
ML	Machine Learning
DL	Deep Learning
SVC	Support Vector Classification
KNN	K-Nearest Neighbor
LR	Logistic Regression
DT	Decision Tree
LDA	Linear Discriminant Analysis
QDA	Quadratic Discriminant Analysis
RF	Random Forest
GB	Gradient Boosting
XGBoost	Extreme Gradient Boosting
Light GBM	Light Gradient Boosting Machine
DANN	Deep Artificial Neural Network
GRU	Gated Recurrent Unit
LSTM	Long Short-Term Memory
AR	Association Rule
CNN	Convolutional Neural Networks
RNN	Recurrent Neural Networks
FDT	Fast Decision Trees
BN	Bayesian Networks
NB	Naïve Bayes
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative

Introduction

Education has recently come to play a vital role in all fields and is indispensable in our lives. The main challenge for universities is students’ failure in courses and also a failure in selecting suitable courses for achieving the best performance. Therefore, a vital factor in determining the accomplishments of any educational institution is the academic performance of the students (Mengash, 2020; Ramaphosa et al., 2018). Selecting suitable courses is considered the main way to avoid failure in a career. Thus, this study will focus on predicting student grades and using recommendation systems to recommend suitable courses for students during academic study (Dwivedi & Roshni, 2017).

Educational Data Mining (EDM) is the use of Data Mining (DM) methods in educational institutions to analyze available data and extract available knowledge. Using EDM, various data mining techniques can be applied, such as classification, Association Rule (AR) mining, and regression, in the education field. EDM is considered the most common technique used in predicting students' grades to extract useful information to help students and educational institutions (Kouser et al., 2020).

Machine Learning (ML) comprises a wide range of algorithms aimed at facilitating computers acquire the ability to recognize patterns within data and subsequently make predictions or decisions without the need for explicit programming. ML encompasses various categories of algorithms, including supervised learning, which entails training models on labeled data for predictive purposes; unsupervised learning, which revolves around uncovering patterns within unlabeled data; and reinforcement learning, which centers on learning optimal actions through interaction with an environment (Alenezi & Faisal, 2020).

In contrast, Deep Learning (DL) represents a sub-field of ML that centers on artificial neural networks inspired by the human brain. These neural networks comprise numerous interconnected layers of neurons, or nodes, enabling them to automatically extract intricate features from data. Convolutional Neural Networks (CNNs) excel in image analysis, Recurrent Neural Networks (RNNs) are well-suited for sequential data, and more advanced architectures like Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) are adept at capturing long-term dependencies within sequences (Veluri, 2022; Kim, 2016).

The fundamental distinctions between ML and DL can be identified in their feature engineering, scalability, and performance characteristics. ML typically necessitates manual feature engineering, whereas DL automates this process by employing deep neural networks (Perrotta & Selwyn, 2020). DL models shine when dealing with extensive datasets and high-dimensional data thanks to their capacity to discern intricate patterns, but this advantage may come with increased computational demands. In contrast, ML models offer greater interpretability and are better suited for smaller datasets (Khokhar & Borst, 2022).

ML and DL play pivotal roles in the field of education by predicting students' academic performance, thereby aiding students in attaining high grades and mitigating academic risks (Adnan, 2021). Predicting students' academic success serves as a critical process for assessing their grades early in a semester. Multiple DM techniques are employed to predict various educational outcomes, including academic performance, dropout rates, and overall success. These techniques enable educational institutions to proactively address challenges and optimize students' learning experiences (Dwivedi & Roshni, 2017). Predicting students' academic performance in the early stages serves as a valuable tool for initiating timely interventions to improve their performance and gain insights into the faculty's failure rate. There are various features that can affect a student's performance, such as academic, non-academic, behavioral, and others. But EDM has important tools to solve and deal with these different features (Alyahyan & Düştegör, 2020).

Recommendation systems are very important in social networking, e-commerce, entertainment, and e-learning. Recommendation systems are very helpful in e-learning to help students select suitable courses (Akhtar, 2020; Grewal & Kaur, 2016). In recent times, recommendation systems have proven their effectiveness in the education sector. These systems have gained popularity for providing a wide range of recommendations, benefiting students, teachers, schools, and various other stakeholders (Alshaikh, 2021; Thanh-Nhan et al., 2016).

In this research, as shown in Fig. 1, the authors present two stages: the first is a prediction model for students' performance, and the second is a recommendation model for suitable departments and courses. In the first stage, (the MU-dataset) was gathered for undergraduate students from Mansoura University. The authors applied some DL and ML algorithms for developing this predictive model to evaluate the students' data to predict students' performance in all courses of the second academic year based on the grades of the previous first academic year. which will help them in choosing the most appropriate courses and departments for them. Ram et al. (2021); Macarini and Antonio (2019); Bennedsen and Caspersen (2019). Then, in the next stage, the authors focused on the recommendation system. This model has been built

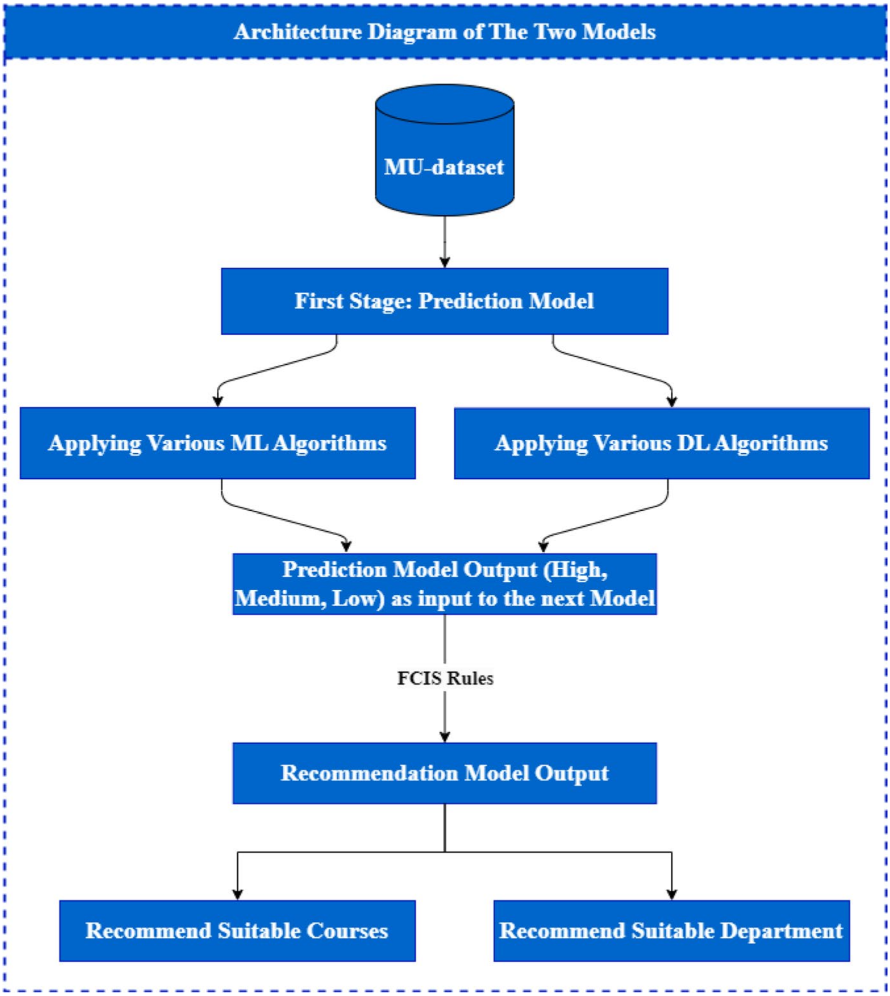

Fig. 1 Architecture diagram of the two models, the first predicts students' performance, and the second recommends the suitable department and courses

based on FCIS rules to give advice to students about enrolling in suitable courses. Consequently, this model is more important for students to make a good selection for suitable courses and is essential to improving their education level. Akhtar (2020); Grewal and Kaur (2016); Obeidat et al. (2019).

as shown from Fig. 1 the recommendation model takes its input from the output of the previous prediction model, produces an output as a list of suitable courses based on the FCIS rules, and recommends a suitable department to avoid failure in courses and fully achieve the best performance.The following are some of this research study's contributions:

1. Without depending on an online dataset, the authors collected a real dataset and performed data pre-processing tasks to achieve better results.
2. Applying data normalization to the variety of independent data features.
3. Applying various validation methods such as stratified 6-fold cross-validation and random hold-out.
4. Applying various ML techniques such as Support Vector Classification (SVC), K-Nearest Neighbor (KNN), Logistic Regression (LR), Decision Tree (DT), Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), Random Forest (RF), Gradient Boosting (GB), Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (Light GBM), and Extra Trees to predict students' performance in three categories Low, Medium, and High.
5. Applying various DL techniques such as Deep Artificial Neural Network (DANN), Gated Recurrent Unit (GRU), and Long Short-Term Memory (LSTM)to predict students' performance in three categories: Low, Medium, and High.
6. Measuring the performance using various evaluation methods, such as F1-Score, recall, accuracy, and precision.
7. Work on comparing with more papers, such as in related work, to get the best results.
8. Applying ML and DL to prediction models (first stage).
9. Applying a recommendation system to recommend suitable courses and choose the best department for students (second stage).

The remaining sections of this paper are written as follows: related work to give an overview of some related works in sect 2. Methods and materials are described in sect 3. Experimental results and discussion to show and discuss the results are shown in sect 4. Finally, conclusions and future work are presented in sect 5.

Related works

One of the most important factors in higher education is students' academic performance (Dwivedi & Roshni, 2017; Al Mayahi & Al-Bahri, 2020). Several researchers (Adnan, 2021; Bennedsen & Caspersen, 2019) used EDM applications to evaluate and predict students' academic performance to understand the learning process. Also, some researchers used EDM applications to work with recommendation systems to make decisions to understand the learning process (Akhtar, 2020; Grewal & Kaur, 2016). The recommendation system benefits undergraduate students who need advice in course selection processes during the pre-registration period in the semester (Lin, 2018). This section will discuss some previous studies on predicting students' academic performance using various ML and DL algorithms and also present research on recommendation systems that were performed based on real and online datasets.

Predicting students' performance

Many studies have been conducted to predict students' academic performance in order to help students in choosing their future career path. According to the researchers in Al Mayahi and Al-Bahri (2020) a real dataset from different faculties was used to predict students' academic performance. They applied SVC and Naïve Bayes (NB) to predict academic students' performance in two categories (Good, Pass). The dataset was collected from the University of Nizwa and consisted of 550 records. It was observed that the SVC algorithm has a higher accuracy rate of 89%.

The researchers in Farissi et al. (2020), and Sixhaxa et al. (2022) used an online dataset gathered from Kaggle; called the Kalboard 360 dataset repository, to measure students' academic performance. There are 480 records in this dataset, with 16 characteristics. The researchers in Farissi et al. (2020) applied various ML techniques such as DT, Artificial Neural Network (ANN), RF, voting, bagging, and boosting to predict grades in three categories (Low, Medium, and High) and observed that GA and RF algorithms had a higher F1-score rate of 81.18%, whereas the researchers in Sixhaxa et al. (2022) applied various ML algorithms such as RF, gaussian NB, SVM, K-NN, and LR to predict grade in three categories (Low, Medium, and High) and observed that the SVC algorithm had a higher accuracy rate of 81.67%.

The researchers in Hussain et al. (2019) used various ML and DL techniques, and the DL achieved good results in these studies. They applied various DL techniques to the real dataset gathered from India's three different faculties to predict academic students' performance in two categories (Pass, and Fail) and observed that the RNN algorithm had a higher accuracy rate of 95.34%.

The researchers in Aggarwal et al. (2021); Liu (2022), and Orji and Vassileva (2022) used a real dataset gathered from different faculties to assess students' academic performance and aimed to reduce dropout levels by helping students' prediction performance in two categories. They used academic and non-academic features with different sizes for the dataset, as shown in Table 1. The researchers in Aggarwal et al. (2021) collected a real dataset from technical institutions in India's Uttar Pradesh state and applied various ML techniques (J48 DT, LR, MLP, SVM, AdaBoost, bagging, RF, voting) and observed that the RF algorithm had a higher F1-score rate of 93.8%. Whereas the researchers in Liu (2022) applied GBT, 1D-CNN, LR, KNN, RF, and LSTM and observed that LSTM algorithm had a higher accuracy rate of 90.25% in two categories (Pass, Fail), whereas the researchers in Orji and Vassileva (2022) applied RF, LR, SVM, DT, and KNN on more

Table 1 Previous works' Comparison in predicting students' academic performance

Paper	Year	Dataset Source	Dataset Size	algorithms	Highest score	Data Mining Task	Output Feature (Classes)
Al Mayahi and Al-Bahri (2020)	2020	Nizwa University, Real Data	550	NB and SVC	SVC achieved 89% accuracy	Classification	2 (Pass, Fail)
Farissi et al. (2020)	2020	Kaggle, Online Data	480	DT, ANN, RF, Voting, Bagging and Boosting	GA and RF achieved 81.18% accuracy	Classification	3 (Low, Medium, High)
Sixhaxa et al. (2022)	2022	Kaggle, Online Data	480	Gaussian NB, SVM, RF, KNN, and LR	SVC achieved 81.67% accuracy	Classification	3 (Low, Medium, High)
Hussain et al. (2019)	2019	different Indian faculties, Real Data	10140	RNN, Artificial Immune Recognition System v2.0 (AIRS2) and AdaBoost	RNN achieved 95.34% accuracy	Classification	2 (Pass, Fail)
Aggarwal et al. (2021)	2021	Indian technical institutions, Real Data	6807	J48 DT, LR, MLP, SVM, AdaBoost, Bagging, RF and Voting	RF achieved 93.8% F1-score	Classification	2 (Complete, Withdraw)
Liu (2022)	2023	the Open University Learning Analytics Dataset (OULAD), Real Data	5341	GBT, 1D-CNN, LR, K-NN, RF, and LSTM	LSTM algorithm achieved 90.25% accuracy	Classification	2 (Pass, Fail)
Orji and Vassileva (2022)	2022	The university dentistry students, Real Data	924	RF, LR, SVM, DT, and KNN	RF achieved 94.9% accuracy	Classification	2 (High, Low)
Kehinde et al. (2022)	2022	Kaggle, UCL, Online Data	396	ANN	ANN achieved 92.3%	Classification	2 (Pass, Fail)
Latif et al. (20232)	2023	Students' Online activities, Online Data	575	RF, SVM, FDT, Bayesian Network (BN), NB, and LR	bagging and boosting FDT achieved 98.25%	Classification	2 (Pass, Fail)
Nabil et al. (2021)	2021	FCIS at Mansoura University, Real Data	4266	DNN, RF, GB, DT, LR, KNN and SVC	DNN achieved 89%	Classification	2 (Pass, Fail)

Table 1 (continued)

Paper	Year	Dataset Source	Dataset Size	algorithms	Highest score	Data Mining Task	Output Feature (Classes)
Ya˘gcı(2022)	2022	University in Turkey, Real Data	1854	RF, NN, SVM, LR, NB and KNN	RF achieved 74.6%	Classification	4
Pujianto et al. (2020)	2020	Kaggle (Kalboard) 360), Online Data	480	KNN and DT	DT achieved 71.09% accuracy	Classification	3 (Low, Middle, High)

experiments and observed that the RF algorithm had a higher accuracy rate of 94.9% in two categories (High, Low).

The researchers in Kehinde et al. (2022), and Latif et al. (20232) used an online dataset gathered from Kaggle and students' online activities. The researchers in Kehinde et al. (2022) used an online dataset consisting of 396 records collected from Kaggle and applied ANN. They observed that the ANN algorithm had a higher accuracy rate of 92.3% in two categories (Pass, and Fail), whereas the researchers in Latif et al. (20232) worked on students' online activities to improve the process of predicting performance and applied more ML techniques such as RF, SVM, Fast Decision Trees (FDT), Bayesian Networks (BN), NB, and Linear Regression (LR) and observed that bagging and boosting FDT algorithms had a higher accuracy rate of 98.25% for binary classification (Pass, and Fail). The researchers in Nabil et al. (2021) used a real dataset collected from FCIS Mansoura University that consisted of 4266 records. They used eleven features of the first level as input to predict the grade of the course of the data structure of the second level in two categories (Pass and Fail). They applied more ML techniques, such as DNN, RF, GB, DT, LR, KNN, and SVC, and observed that the DNN algorithm had a higher accuracy rate of 89%.

The researchers in Ya˘gcı (2022) used a real dataset gathered from a university in Turkey that consisted of 1854 records. They applied more ML techniques, such as RF, ANN, SVM, LR, NB, and KNN, and observed that the RF algorithm had a higher accuracy rate of 74.6% in four class labels. The researchers in Pujianto et al. (2020) used an online dataset gathered from Kaggle and students' online activities. They aimed to lower dropout rates by assisting students in predicting their grades in a course before enrolling in it. They used an online dataset gathered from Kaggle. They applied KNN and DT and observed that the DT algorithm had a higher accuracy rate of 71.09% in three categories (Low, Middle, and High). Table 1 compares some of the previous studies which predict students' academic performance.

Recommending students' academic courses

A recommender model was developed in Bydžovská (2016) to help in student enrollment decisions. This model may be more appropriate because it has a method for recommending students to elective courses depending on their interests, knowledge, free time, and skills. This model was built on four algorithms: the first looked for the most common and enrolled courses; the second discovered similarities between courses based on the interests of each student; the third recommended courses taught by students' favorite teachers; and the fourth algorithm calculated social ties among students and chose courses that they were interested in. The first algorithm was discovered to be the most appropriate for optional course recommendations.

A recommender system was developed in Bhumichitr et al. (2017) to assist undergraduate students. The collaboration-based recommendation employing Alternating Least Squares algorithms (ALS) and the Pearson correlation coefficient were utilized in this model, which was based on the similarity between the students' course templates. A real dataset was used in this model. The dataset contained information from 3,614 students who had completed 52 courses over the previous 8 years. The accuracy rate of applying ALS algorithms was 86%.

This recommendation algorithm was created by Chen and Deng (2021) to identify students' interests based on the learning behaviors in the network and to provide video learning materials that are related to those interests. They used the Collaborative Filtering (CF) technique based on items in ML and the Pearson correlation coefficient method to find extremely similar video materials and then recommended the learning resources they were interested in. The accuracy was found to be high.

A recommender model was developed by Akhtar (2020) to recommend suitable courses in Pakistan's universities. The grades of students were the most important factor in this model. This model is important in determining the CGPA and was used to calculate expected marks and grades using collaborative filtering (CF) and a rating prediction approach. The dataset was tested using simulated data from 2,600 students and 470 currently available courses. The accuracy was found to be within an acceptable range.

The researchers in Esteban et al. (2020) developed a recommender system as a tool to help students choose appropriate courses based on their personal interests and academic performance. This model was based on a hybrid recommendation system that used content-based filtering and CF and was influenced by factors (e.g., courses and student information). The dataset was real and came from the University of Cordoba's Computer Science degree in Spain. The dataset included three academic years, 95 students, 63 courses, and 2,500 entries. The hybrid system was discovered to be more appropriate than others.

The researchers in Rahman et al. (2022) developed a recommendation system as a tool to help students select suitable courses during their undergraduate studies based on their interests (academic performances and personal interests). The applied ML techniques, such as K-Means clustering to determine the most and least demanded courses among students and FP-Growth to generate the rules to recommend appropriate courses, used a real dataset that was collected from the computer science department.

Materials and methods

This research has proposed a prediction and then a recommendation as two models in one system. The main goal is to build a predictive model using ML and DL techniques to predict the academic performance of students in upcoming courses in advance at the next level based on their previous academic degrees, then use the previous model as input to the recommendation model to recommend courses that are commensurate with students' skills, behaviors, and hobbies, which helps students get good grades during their studies and also recommends the most suitable department for them. These two models will be a grateful gift to students because they will know their probable grades for the next year at an early stage and will also know their suitable department.

The study process

The process of our study consisted of five stages. First stage: data collection. Second stage: data pre-processing and cleaning up noise and missing values before entering the model. In the third stage, the data will be the input of the prediction model after pre-processing to early predict students' grades (performances). Fourth stage: model evaluation, testing, showing results, and comparison with other results, as declared in related work. Finally, the recommendation system that was built by the previous model of the previous stage will be the input for it to recommend the top courses for the students in the next level and the best department for you. The prediction results were obtained by using eleven ML techniques and three DL techniques that will be listed in detail.

Instruments and tools

The development process of this study has utilized the Python programming language and used some notebooks and editors to execute the experiment, such as Jupyter Notebook, Google Kollab Notebook, and Spyder Editor. On the other hand, it has also used the Scikitlearn Python library for the ML algorithms and the TensorFlow Python library for the ML algorithms and DL.

Datasets collection

This work has depended on real data for the created dataset. The authors of this paper gathered the (MU-dataset) for undergraduate students over the twelve-year span from 2008 to 2020 at Mansoura University's Faculty of Computer and Information Sciences. The dataset of the current study included 3981 records with 11 features, representing all students' degrees in all courses they took in the first and second academic years. These features are purely academic and have to do with the degrees that students get within the first academic year. The (MU-dataset) is presented in Table 2.

The (MU-dataset) contains eleven academic features. These features are the only ones that were used because the university gathered only academic features. These features have been used to build two types of models: predictive and recommendation models. These two models will be used in one system and can be used in universities to cut down on failures. This system is considered very important for many universities because there are many universities that gather only academic features like ours. In the first academic year, the first semester includes five courses: physics, computer science, discrete mathematics, English, and the fundamentals of programming. Then, the second semester includes six courses: Logic, Integrals and Derivatives, Information System, Probability I, Information Technology, and Object-Oriented Programming. The full mark for each course is 100, and the student must get at least 50 marks to succeed in each course. The features of the dataset are listed in Table 2. The first 11 rows from table number

Table 2 MU-Dataset Description

number	Feature Name (input)	Type	Feature Value Range	Feature Description
1	Physics	Number	From 0 to 100	Physics Course
2	CS	Number	From 0 to 100	Computer Science Course
3	D-Math	Number	From 0 to 100	Discrete Mathematics Course
4	E	Number	From 0 to 100	English Course
5	Fund-Prog	Number	From 0 to 100	Fundamental of Programming Course
6	Logic	Number	From 0 to 100	Logic Course
7	Integral	Number	From 0 to 100	Integral and Derivative Course
8	IS	Number	From 0 to 100	Information System Course
9	Prob1	Number	From 0 to 100	Probability 1 Course
10	IT	Number	From 0 to 100	Information Technology Course
11	OOP	Number	From 0 to 100	Object Oriented Programming Course
12	target_web	Ordinal	Low, Medium, High	Web Design Course
13	target_ds	Ordinal	Low, Medium, High	Data Structures Course
14	target_os	Ordinal	Low, Medium, High	Operating System Course
15	target_linear	Ordinal	Low, Medium, High	Linear Algebra Course
16	target_prob2	Ordinal	Low, Medium, High	Probability 2 Course
17	target_arch	Ordinal	Low, Medium, High	Computer Architecture and Organization Course
18	target_dc	Ordinal	Low, Medium, High	Data Communication Course
19	target_graphic	Ordinal	Low, Medium, High	Computer Graphics Course
20	target_db	Ordinal	Low, Medium, High	Database Course

represent the 11 course degrees in the first academic year (first two-semester course degrees). From row numbers 12 to 20, considered the target feature, these nine features represent the nine courses' grades for the second academic year. The model will use only one target at a time. Figs. 2 and 3 display a sample of the input feature and the target feature (the class label of the feature), respectively.

In this work, the authors' goal is to predict the courses grades of all students as [Low, Medium, High] in the second academic year(third and fourth semesters) and then use these predicted grades in the second model (recommendation model) to rank the suitable courses for the students that enroll in them and also select the suitable department for them. Fig. 4 presents the architecture diagram of the proposed approach.

	A	B	C	D	E	F	G	H	I	J	K
1	Physics	CS	D-Math	E	Fund-Pr	Logic	Integral	IS	Prob1	IT	OOP
2	48	55	21	62	58	50	66	61	41	74	63
3	64	60	65	62	54	64	57	57	55	54	62
4	69	60	56	71	54	32	73	50	35	58	49
5	38	46	51	61	45	57	67	42	41	57	52
6	64	65	56	79	38	25	64	52	36	53	42

Fig. 2 Sample of the input features (The first academic year courses)

L	M	N	O	P	Q	R	S	T
target_web	target_ds	target_os	target_linear	target_prob2	target_arch	target_dc	target_graphic	target_db
Medium	High	Low	Medium	High	Medium	High	Low	Medium
High	High	Medium	Low	High	Medium	Medium	Low	Medium
Low	Low	High	Medium	High	Low	Medium	Medium	Low
Low	Medium	Low	High	Medium	Medium	High	Medium	High
Medium	Low	High	High	Low	High	Medium	Medium	High

Fig. 3 Sample of the target features (The second academic year courses))

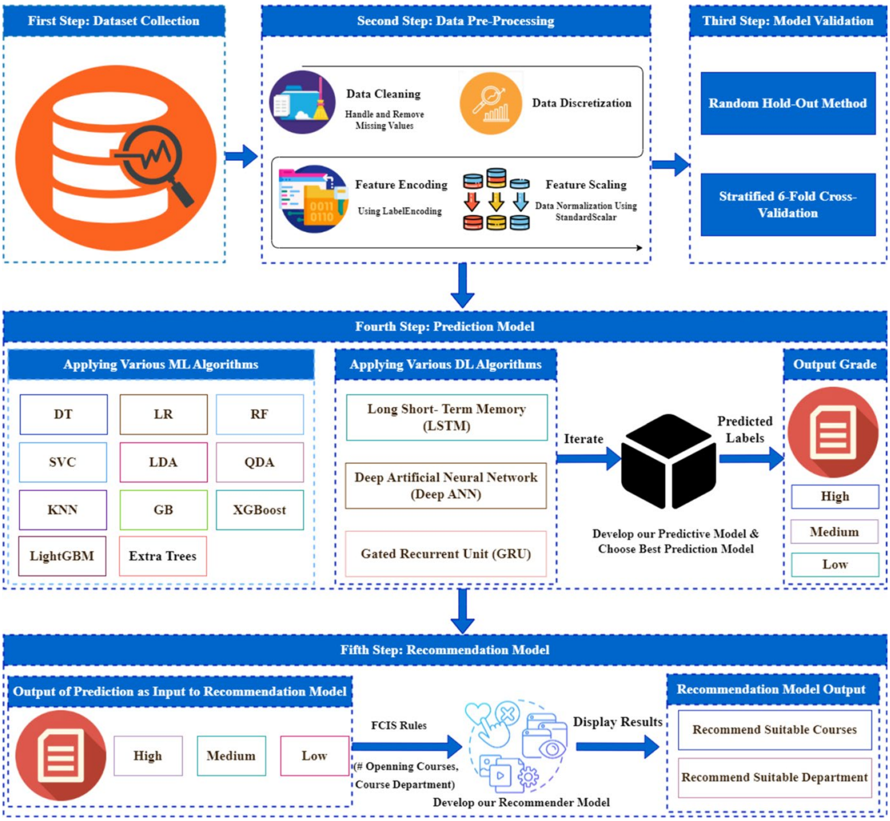

Fig. 4 The proposed model framework's architecture

Data pre-processing

It is a crucial and one of the most significant processes in DM that improves the data quality and guarantees a more effective and efficient modeling process (Hassan et al., 2020). In the pre-processing steps, the raw data are transformed into the proper format to solve the issues and errors in the real dataset collected from the real world (Kotsiantis et al., 2006; Ghorbani & Ghousi, 2020). This process contains the following steps: data cleaning, data discretization, feature encoding, and data normalization (data scaling). The details of those actions are described below.

Data cleaning

The cleaning process is the process of eliminating inconsistencies or null records, detecting outliers, removing noise, and completing missing data. Nabil et al. (2021). Data cleaning must be realized before data analysis. In this study's real dataset, there are some missing values, such as the student's absence from any exam or some records dropped by failure in an Excel CSV file, etc. So, authors need to apply a cleaning process to their work to fix these problems.

Handling missing values

To handle the missing values in the study dataset, several preliminary experiments have been conducted with various imputation methods, including median imputation, removing rows with missing values, and substituting a constant value. However, after evaluating the performance of these approaches, we found that mean imputation provided the most accurate and consistent results in our model. Given its effectiveness, the mean imputation has been applied as the study's primary method for handling missing values, balancing simplicity with performance. This decision ensures that the study dataset maintains integrity and reduces potential biases that missing data introduces.

Handling outliers

Regarding outliers, students' grades have been classified into three ranges: (0–50), (50–80), and (80–100). No student can have a grade less than zero or greater than 100. Therefore, there is no need to apply outlier handling techniques in this study. On the other hand, the study dataset is based on actual student grade distributions, which naturally include variations that reflect real academic performance. By retaining this data as-is, we aim to preserve the authenticity of the dataset, believing it represents real-world scenarios that our model is designed to predict. Adjusting students' scores might inadvertently distort these natural variations, potentially limiting the model's applicability to genuine academic records.

Data discretization

One of the pre-processing steps that the authors have found to be the most effective is discretization (Kohavi & Sahami, 1996). The numerical dataset used in this study is used to predict students' performance, where students' marks in the first two years should be considered to predict future students' performance (Kotsiantis et al., 2006; Grandgirard, 2002). In this study, in order to represent the class labels for the classification problem, the numerical values of students' degrees have been converted into ordinal values using the discretization mechanism

(Pfahringer, 1995). Many recent studies (Farissi et al., 2020; Sixhaxa et al., 2022), and Pujianto et al. (2020) have classified the datasets into three classes (low, medium, and high) for many reasons such as balancing the dataset. If the data is imbalanced such as (more students in the middle range), categorizing into these ranges can help balance the classes, making it easier for the model to learn and generalize from the data. Table 3 displays the input feature as numerical and categorizes the target (class label) into three ordinal classes (Low, Medium, and High) based on student's grades. These ranges (0–49: low, 50–79: medium, and 80–100: high) have been established by the university. A student who scores less than 50 in a course is considered a failing student, while a student who scores 50 or above is considered successful. We have further categorized the 50–100 range into two classes: (80–100) in the high range, and (50–79) in the medium range.

Feature encoding

This stage is also very important when using machines and DL techniques. The feature encoding has been used to convert categorical data into numerical data because machine and DL algorithms cannot work with categorical data (Nabil et al., 2021). In order to feed the input features into the model, these categorical features were converted into numerical form. In this work, a Python library called LabelEncoder has been utilized. Using this encoding method, each ordinal variable is assigned an integer value, and the categorical data is transformed into a numerical form. In the (MU-dataset), input features are integer numbers, but the output features were converted into integer numbers as High is 0, Low (Fail) is 1, and Medium is 2.

Data normalization (feature scaling)

Data normalization, or feature scaling, is an essential method that aids in normalizing the dataset's range of independent features. This stage accelerates the learning process of the algorithm (Ghorbani & Ghousi, 2020; Amrieh et al., 2016). Without this step, the accuracy decreased, which also affected the efficiency of the algorithm. So, the authors used a method that is widely used in normalization, called a standard scaler, to get the most efficient way to apply algorithms and achieve higher accuracy. In this work, μ and σ for that feature have been computed to standardize each input

Table 3 Discretization step results

Feature Shape Input feature	Class labels (Three intervals) Student marks (Degrees)	Class labels
From 0 to 100	From 0 to 49	Low (Fail)
	From 50 to 79	Medium
	From 80 to 100	High

variable's feature before entering the algorithm. Then, the following formula is used to determine the new value of Xscaled for each value of X:

$$X(\text{ Value Scaled }) = \frac{x - \mu}{\sigma} \tag{1}$$

where μ and σ are the mean and standard deviation, respectively.

Model validation

In this section, we talked about the techniques used in splitting the (MU-dataset). A validation technique called cross-validation is used to evaluate how well the results of statistical analysis generalize to an independent dataset (Bro, 2008; Schaffer, 1993). In this research, the two most popular approaches to cross-validation are used: six-fold cross-validation and random hold-out. The next subsection will briefly describe these two approaches:

Random hold-out method

Random holdout is a strategy frequently employed in the fields of machine learning and statistical modeling to evaluate the effectiveness of a model. It involves the random selection and removal of a portion of the dataset from the training set, which is then reserved for validation or testing purposes. The model is trained using the remaining data, and its performance is subsequently assessed on the holdout set (Murphy, 2012).In this study, The dataset is randomly divided into 80% as a training set and 20% as a testing set.

K-fold cross-validation

Splitting the dataset into training and testing sets is another technique from cross-validation techniques. The dataset is randomly divided into K equal-sized subsets using this method (Ghorbani & Ghousi, 2020; Kumari et al., 2018). During the building of the model, one of these subsets was used for the testing set, and the other subset was used for the training set.

Stratified K-fold is a method of K-fold cross-validation that was used in this work. In order to train the classification algorithms on the (MU-dataset), six-fold cross-validation was used. The dataset was divided using this method into six equal subsets, one of these subsets is used for training and the others are utilized for testing. The average error rate on test examples is computed as the result of repeating this process six times. First, the classification model was trained. The validation process then begins. The validation phase is the last phase of building a predictive model and is used to evaluate the prediction model's performance (Kumari et al., 2018).

Applying machine learning and deep learning techniques

In this study, various supervised and unsupervised methods were employed to extract hidden patterns within datasets and subsequently classify the data. The authors utilized a supervised machine learning approach besides some DL techniques for classification, with three classes (High, Medium, and Low) for all attributes (Engr, 2020; Romero et al., 2008). Precisely, we applied eleven ML algorithms and three DL algorithms.

Machine learning models

In this study, algorithms have been utilized for both binary classification tasks involving two classes and multi-class classification problems with more than two classes utilized by the authors. These algorithms have been used as follows:

(A) DT constitutes a modeling algorithm based on classification partitioning, and it finds significant application, particularly in the field of EDM (Liakos, 2018; Mahesh, 2020). This method serves the purpose of approximating a valued target function, which is subsequently represented in the form of a DT. The procedure involves categorizing values by arranging them from the root to specific leaf nodes, which are determined by the feature values. Each branch signifies a potential value for a given feature, while each node represents a test condition on an instance's attribute. The initiation of the instance classification process commences at the root node. The classification process, for instance, commences at the root node and proceeds down the tree along edges corresponding to the output feature test values. Depending on the node's value, it proceeds within the sub-tree headed by the new node connected to the previous one. The final classification category or the final class label decision is determined by the leaf node. In this algorithm, various statistical measures, including the Gini index, gain, entropy, and Chi-square, are computed for each node to assess the node's significance (Alzubi et al., 2018; Du & Zhan, 2002). (B) LR is the most common technique used to categorize target variables. This technique is considered the most famous parametric method. In general, this model is a linear approach utilized to predict a categorical dependent variable by considering various independent predictor values (Liakos, 2018; Alzubi et al., 2018). (C) RF stands out as the prevailing ensemble learning technique applied in both regression and classification. This method employs a bagging approach to generate a collection of DTs, each trained on a random subset of the data. The RF combines the output from this ensemble of trees to produce the final decision. The RF algorithm comprises two distinct phases: the first phase involves creating the random forest, while the second phase entails making the ultimate prediction using the random forest classifier established in the initial phase. (D) SVC is the most common technique among the SVM algorithms utilized for classification problems. This algorithm operates within the framework of supervised learning. Within this algorithm, each data item is treated as a point residing in an n-dimensional space, where 'n' corresponds to the number of attributes or features present in the dataset. The SVC accomplishes the classification of data values into distinct class labels by identifying a hyperplane that effectively separates the target training datasets. Its operation centers around maximizing the distances between the nearest data points (Alzubi et al., 2018). (E) LDA is a pivotal classification method in probabilistic and statistical learning. It can be employed for supervised dimensionality reduction by projecting the input features into a linear subspace defined by directions that optimize the distinction between classes. The output dimension must be less than the number of classes. In this work, we used it to classify target features (output features) into three classes (High, Medium, and Low) (Ghojogh & Crowley, 2019). (F) QDA is also a fundamental classification method in probabilistic and statistical learning, like the previous technique (LDA). This algorithm is used for solving binary classification problems (two classes) as well as multi-class classification problems (more than two classes). In this work, we used it to classify the target feature (output feature) into three classes (High, Medium, and Low) (Ghojogh & Crowley, 2019). (G) KNN ranks among the most commonly employed non-parametric techniques in both regression and classification. This supervised machine learning algorithm relies on the value of 'K', which represents the number of nearest neighbors utilized by the classifier to formulate its predictions. In the KNN algorithm, it calculates the distance between the class label value (record) and all the training data records. Subsequently, it identifies the K closest data records within the training dataset. The computation of this distance can take various forms, including Euclidean distance, Manhattan distance, Hamming distance, Minkowski distance, and Chebyshev distance (Romero et al., 2008; Liakos, 2018; Alzubi et al., 2018; Wang et al., 2007). (H) GB is a powerful ML technique that is used in prediction. The additive modeling problem can also be solved using this type of algorithm. The objective of the GB technique is to discover an additive model that minimizes the loss function by employing a numerical optimization algorithm (Hassan et al., 2020; Touzani et al., 2018; Natekin & Knoll, 2013). (I) XGBoost stands as a widely embraced prediction algorithm integrated into numerous machine learning tasks as an integral component of ensemble methods. XGBoost leverages the ensemble method, specifically boosting, to achieve improved predictive performance with class-based data (Hanif, 2020; Freund et al., 1999), (J) LightGBM represents a gradient learning approach that incorporates elements from DT and certain boosting concepts (Fan, 2019). This technique differs from XGBoost in that it uses histogram-based algorithms to get a fast training process and reduce memory consumption. It is also considered in the main prediction algorithm used in classification. (K) Extra Tree is one of the classification techniques used in prediction. It is easily converted to if-then rules, has explicit meaning, and has simple properties. The extra tree is also used for randomizing properties for inputs. This concept proves highly advantageous in scenarios that entail a substantial volume of numerical features (Sharaff & Gupta, 2019).

In this research, all the above techniques (ML models) have been documented in Table 4, along with their respective parameter settings.

Deep learning models

In this work, Our main goal was to comprehensively compare different models, including traditional machine learning methods and more complex deep learning architectures, to identify the most effective approach for the study prediction model. The study underscores the importance of empirical testing to determine the best approach for the study problem and dataset. So, we chose to include DNN LSTM and GRU architectures alongside traditional machine learning algorithms to explore the adaptability and performance

Table 4 The specific parameters and settings for the machine learning techniques (models)

#Number	ML Models	Parameters
1	DT	Criterion='entropy', splitter='best'
2	LR	Penalty='l2', C=1.0, solver='sag'
3	RF	Criterion = 'gini', n_estimators=200, max_depth=11
4	SVC	C=3, kernel='rbf', degree=1, tol=0.001, cache_size=200, decision_function_shape='ovr', verbose=False, probability=False
6	QDA	Reg_param=0.0, tol=0.0001, store_covariance=False
7	KNN	N_neighbors=7, leaf_size=30, weights='distance'
8	GB	n_estimators=100, max_depth=3
9	XGBoost	Objective='binary:logistic', max_depth=5, n_estimators=150, learning_rate=0.5
10	LightGBM	Learning_rate=1
11	Extra Trees	No Parameters

of these sequential models on tabular data. Recent studies (Hussain et al., 2019; Liu, 2022) have shown their potential in capturing complex patterns even in non-sequential, and tabular datasets. They can automatically learn and create new feature representations during the training phase. This reduces the need for manual feature extraction, which is often required for traditional machine learning methods. Including LSTM and GRU in the study helped us highlight the versatility of these models and provided a broader perspective on their applicability. In this study, we applied the two most effective DL models suitable for the numerical dataset (DNN, and RNN(LSTM, GRU)). These algorithms are used for solving binary classification problems (two classes) as well as multi-class classification problems (more than two classes).

Firstly, D-ANN, or DNN, represents one of the most prevalent and versatile deep learning techniques. This method is designed to mimic certain functions performed by the human brain and has demonstrated success in addressing a broad spectrum of DL problems. Often regarded as the most potent variant of ANN (Mohammadhassani, 2013; Zhang et al., 1998; Jain et al., 1996), DNN excels at modeling and solving real-world nonlinear function challenges. The core architecture of this algorithm consists of multiple layers, encompassing more than just a single layer, with each layer comprising processing units known as neurons (artificial neurons). These neurons communicate via connection links, transmitting signals between them. Each connection link is associated with a weight that scales the transmitted signal, and each neuron applies an activation function to the weighted sum of input signals, ultimately producing an output signal (Li et al., 2017). One of the primary advantages of deep ANN is its capacity to discover hidden patterns and useful knowledge from the dataset. In the context of this research, DNN was leveraged for the prediction of students' academic performance. The proposed model is structured with one input layer, twelve hidden layers, and one output layer. The configuration parameters for the DNN model are provided in Table 5.

The model comprises eleven neurons in the input layer and three neurons in the output layer (multi-classification problem). Two activation functions have been employed:

Table 5 The configuration parameters for the DNN

Layer Number	Number of Units	Activation Function
Layer 1 (Input Layer)	11	Relu
Layer 2 (1st Hidden Layer)	512	Relu
Layer 3 (2nd Hidden Layer)	512	Relu
Layer 4 (3rd Hidden Layer)	128	Relu
Layer 5 (4th Hidden Layer)	512	Relu
Layer 6 (5th Hidden Layer)	512	Relu
Layer 7 (6th Hidden Layer)	128	Relu
Layer 8 (7th Hidden Layer)	512	Relu
Layer 9 (8th Hidden Layer)	512	Relu
Layer 10 (9th Hidden Layer)	128	Relu
Layer 11 (10th Hidden Layer)	512	Relu
Layer 12 (11th Hidden Layer)	512	Relu
Layer 13 (12th Hidden Layer)	128	Relu
Layer 14 (Output Layer)	3	Softmax

the Rectified Linear Unit (ReLU) for the twelve hidden layers and the SoftMax function for the output layer. Furthermore, the model utilized the sparse_categorical_crossentropy loss function for multi-class classification, and it employed the Adam optimization algorithm, which is based on adaptive moment estimation, to compute errors. The batch size was set to 32, and the training process involved 100 epochs.

Secondly, RNNs are a type of neural network that is suited to processing time series data and other sequential data. In our work, we present an extension to feedforward networks called RNNs in order to allow the processing of variable-length (infinite-length) sequences (Zhao, 2017; DiPietro & Hager, 2020). Here we talk about two of the most popular recurrent architectures in use: LSTM and GRUs.

(A) LSTM is a special type of RNN. LSTMs operate by treating the hidden layer as a memory unit and are adept at managing both Long-Term Memory (LTM) and Short-Term Memory (STM), making them well-suited for efficient and effective computations. LSTMs employ the concept of gates, which includes the forget gate, learn gate, remember gate, and use gate. The forget gate enables the LTM to discard information that is deemed irrelevant or unimportant. Learn Gate combines STM with the current input when necessary, facilitating the application of recently acquired knowledge from STM to the current input. The remember gate merges the STM with the current input, effectively updating the LTM with new information. The use gate leverages LTM, STM, and the current input to predict the output for the current input, thereby updating the STM (Zhao, 2017; DiPietro & Hager, 2020). In this work, the authors applied an LSTM to predict students' academic performance. This model consists of one input layer, seven hidden layers, and one output layer. The specific configuration parameters for our Deep LSTM model are provided in Table 6.

. (B) GRU is the newer generation of RNNs. It is pretty similar to the previous technique (LSTM). GRU used the hidden state to transfer information and mop up (get rid of) the cell state. GRU architecture indeed employs two primary gates: the reset gate and the

Table 6 The configuration parameters for the LSTM

Layer Number	Number of Units	Activation Function
Layer 1 (Input Layer)	11	Relu
Layer 2 (1st Hidden Layer)	200	Relu
Layer 3 (2nd Hidden Layer)	200	Relu
Layer 4 (3rd Hidden Layer)	150	Relu
Layer 5 (4th Hidden Layer)	150	Relu
Layer 6 (5th Hidden Layer)	100	Relu
Layer 7 (Output Layer)	3	Softmax

update gate. The reset gate determines how much of the previous state information should be forgotten or reset, allowing the model to selectively forget irrelevant or less important information. The update gate operates in a manner akin to the forget and input gates in the LSTM network. It decides which information from the previous state to discard and what new information should be incorporated into the current state. GRU is a little faster to train than LSTM. GRU and LSTM are very similar, and there isn't a clear winner as to which one is better (Li et al., 2020; Jiao et al., 2020). In this research, the authors utilized a GRU to predict students' academic performance. The architecture of this proposed model comprises one input layer, eight hidden layers, and one output layer. Detailed configuration parameters for the Deep GRU model can be found in Table 7.

The LSTM and GRU models comprise eleven neurons in the input layer and three neurons in the output layer (multi-classification problem). Two activation functions have been employed: ReLU for five hidden layers and the SoftMax function for the output layer. Furthermore, the models utilized the sparse_categorical_cross-entropy loss function for multi-class classification, and they employed the RMSprop optimization algorithm, which refers to Root Mean Squared Propagation, to compute the errors. The batch size was set to 128, and 32 for the LSTM, and the GRU model, respectively. While the training process for both models involved 100 epochs.

Table 7 The configuration parameters for the GRU

Layer Number	Number of Units	Activation Function
Layer 1 (Input Layer)	11	Relu
Layer 2 (1st Hidden Layer)	512	Relu
Layer 3 (2nd Hidden Layer)	512	Relu
Layer 4 (3rd Hidden Layer)	256	Relu
Layer 5 (4th Hidden Layer)	256	Relu
Layer 6 (5th Hidden Layer)	128	Relu
Layer 7 (6th Hidden Layer)	128	Relu
Layer 8 (7th Hidden Layer)	64	Relu
Layer 9 (8th Hidden Layer)	32	Relu
Layer 10 (Output Layer)	3	Softmax

Model evaluation

The authors used four different evaluation measures to evaluate the classification quality namely: F1-score, precision, recall, and accuracy. The Confusion Matrix (CM) shown in Table 8 can be used to calculate the previous measurements.

In much earlier research, accuracy was the performance metric that was most commonly utilized. If there are the same number of instances for each class in the dataset, accuracy can be a useful metric in this case. If not, accuracy serves no use because it predicts the value of the majority class, rendering it useless (Engr, 2020; Khattab et al., 2023). The F1-score is regarded as the average recall and precision value and is particularly useful when there are several class distributions since it contains essential and important results about how well classifiers performed in each class (Chui, 2020; Hasan, 2020; Zaffar, 2021). The following is a representation of the evaluation measures equation used in this study:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{2}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{4}$$

$$\text{F1-Score} = \frac{2TP}{2TP + FP + FN} = 2 * \left(\frac{\text{Precision} \, * \, \text{Recall}}{\text{Precision} \, + \, \text{Recall}}\right) \tag{5}$$

Experimental results and discussion

In this section, we provide a comprehensive overview of the experimental phase of the research study.

Environment

In this study, the authors run this work (experiments) on 16 GB of RAM and a Core i7 processor. We used the Python language and some notebooks and editors

Table 8 Confusion Matrix (CM) representation

		The actual class	
		Positive	Negative
Prediction Condition	Positive	True positive (TP)	False Negative (FN)
	Negative	False Positive (FP)	True Negative (TN)

to execute the experiment, such as Jupyter Notebook, Google Kollab Notebook, and Spyder Editor. We also used the Scikitlearn Python library for the ML algorithms, the TensorFlow Python library for the ML algorithms, and DL. In this study, the authors employed two validation techniques, namely stratified six-fold cross-validation and random hold-out, to divide this dataset into training and testing sets.

Models experiments

ML and DL techniques were applied in two ways (stratified six-fold cross-validation and random hold-out method) as follows:

Results of the random hold-out method (80% training and 20% testing)

The performance of the study models was assessed using four commonly employed evaluation metrics: F1-score, precision, recall, and accuracy. The subsequent subsections will provide a summary of the performance of both the ML algorithms and the DL algorithms.

(A) The performance of the ML algorithms

Fig. 5 presents a comparison of the results of the eleven ML algorithms used.

According to Fig. 5, the highest accuracy is achieved by the SVC, with an accuracy rate of 78.04%. On the other hand, the DT model performs the least satisfactorily, with an accuracy of 61.43%.

Furth

(B) The performance of the DL algorithms

ML Model	*Test Accuracy*	*Precision*	*Recall*	*F1-Score*
DT	61.43%	61.54%	61.43%	60.34%
LR	76.28%	75.03%	76.28%	73.63%
SVC	78.04%	77.20%	78.04%	75.37%
LDA	76.53%	75.31%	76.53%	73.60%
QDA	73.65%	71.70%	73.65%	72.25%
KNN	74.78%	72.56%	74.78%	72.26%
RF	77.54%	76.68%	77.54%	75.05%
GB	75.78%	74.23%	75.78%	73.73%
XGBoost	74.15%	72.78%	74.15%	72.93%
LightGBM	73.90%	72.04%	73.90%	72.36%
Extra Trees	76.28%	74.79%	76.28%	73.51%

Fig. 5 Heatmap for an evaluation of the performance of ML algorithms based on the random hold-out method

DL Model	Test Accuracy	Precision	Recall	F1-Score
DANN	71.14%	70.62%	71.14%	70.55%
LSTM	76.16%	74.84%	76.16%	74.62%
GRU	74.40%	72.40%	74.40%	72.65%

Fig. 6 Heatmap for an evaluation of the performance of DL algorithms based on the random hold-out method

ermore, SVC exhibits notably high precision, recall, and F1-score, with percentages of 77.20%, 78.04%, and 75.37%, respectively. Conversely, the DT model demonstrates the lowest precision, recall, and F1-score, with percentages of 61.54%, 61.43%, and 60.34%, respectively.

Fig. 6 presents a comparison of the results of the three DL algorithms used.

According to Fig. 6, the highest accuracy is achieved by the LSTM, with an accuracy rate of 76.16 %. On the other hand, the DANN model performs the least satisfactorily, with an accuracy of 71.14%.

Furthermore, LSTM exhibits notably high precision, recall, and F1-score, with percentages of 74.84%, 76.16%, and 74.62%, respectively. Conversely, the DANN model demonstrates the lowest precision, recall, and F1-score, with percentages of 70.62%, 71.14%, and 70.55%, respectively.

Results of the stratified 6-fold cross-validation method

(A) The performance of the ML algorithms

ML Model	Test Accuracy	Precision	Recall	F1-Score
DT	61.68%	62.40%	61.68%	62.02%
LR	74.96%	73.10%	74.96%	72.18%
SVC	76.16%	74.81%	76.16%	73.35%
LDA	74.50%	72.36%	74.50%	71.59%
QDA	69.68%	67.65%	69.68%	68.13%
KNN	74.05%	71.55%	74.05%	71.23%
RF	75.11%	73.46%	75.11%	72.23%
GB	77.22%	76.01%	77.22%	75.20%
XGBoost	76.31%	74.93%	76.31%	74.77%
LightGBM	72.24%	70.38%	72.24%	70.87%
Extra Trees	75.86%	74.52%	75.86%	72.40%

Fig. 7 Heatmap for an evaluation of the performance of ML algorithms based on the 6-fold cross-validation

Fig. 7 provides a comprehensive comparison of the performance of the eleven ML algorithms.

According to Fig. 7, the highest accuracy is achieved by the GB, with an accuracy rate of 77.22%. On the other hand, the DT model performs the least satisfactorily, with an accuracy of 61.68%.

Furthermore, GB exhibits notably high precision, recall, and F1-score, with percentages of 76.01%, 77.22%, and 75.20%, respectively. Conversely, the DT model demonstrates the lowest precision, recall, and F1-score, with percentages of 62.40%, 61.68%, and 62.02%, respectively.

From previous results, it can be observed that the SVC algorithm had the highest measures in the random hold-out method, GB had the highest measures in the stratified 6-fold cross-validation method, and DT had the worst measures in both the stratified 6-fold cross-validation method and the random hold-out method.

(B) The performance of the DL algorithms

Fig. 8 presents a comparison of the results of the three DL algorithms used.

According to Fig. 8, the highest accuracy is achieved by the GRU, with an accuracy rate of 76.16%. On the other hand, the LSTM model performs the least satisfactorily, with an accuracy of 74.50%.

Furthermore, GRU exhibits notably high precision, recall, and F1-score, with percentages of 74.73%, 76.16%, and 74.74%, respectively. Conversely, the LSTM model demonstrates the lowest precision, recall, and F1-score, with percentages of 73.48%, 74.50%, and 72.56%, respectively.

From previous results in the stratified 6-fold cross-validation method, it can be concluded that the GRU algorithm has the highest measures and LSTM has the worst measures. There is a difference in measures between the DL algorithm in both the stratified 6-fold cross-validation method and the random hold-out method.

Class imbalance is a well-recognized issue in machine learning that can skew predictions and lead to biased models. Several important steps were taken in this study to address the issue of imbalance and ensure that model performance was evaluated fairly and comprehensively.

Comprehensive evaluation metrics Evaluation metrics less sensitive to class imbalance, such as precision, recall, and the F1-score, were used. These metrics provide a more detailed understanding of model performance, particularly for the minority class, which is often the class of interest in imbalanced datasets. By using these metrics, model performance was evaluated across all classes, rather

DL Model	*Test Accuracy*	*Precision*	*Recall*	*F1-Score*
DANN	75.41%	73.72%	75.41%	71.59%
LSTM	74.50%	73.48%	74.50%	72.56%
GRU	76.16%	74.73%	76.16%	74.74%

Fig. 8 Heatmap for an evaluation of the performance of DL algorithms based on the 6-fold cross-validation

than relying solely on accuracy, which can be misleading when class distribution is uneven.

Classifier-specific robustness: Some of the classifiers used in the study, such as Random Forests and Decision Trees, possess inherent robustness to class imbalance. These models construct multiple decision boundaries and are less likely to be heavily biased toward the majority class. While class imbalance can still impact their performance, these classifiers generally handle it better compared to other models, providing a more balanced view of performance.

Cross-validation: Cross-validation was employed to ensure the model's robustness and to minimize overfitting to any particular subset of the data. This approach helped ensure that the evaluation was more generalizable and not overly influenced by a skewed class distribution in any specific training fold.

In conclusion, the class imbalance issue was addressed through a comprehensive approach. Robust evaluation metrics like precision, recall, and F1-score were used, the inherent strengths of classifiers like Random Forests and Decision Trees were leveraged, and cross-validation was incorporated to ensure generalizability.

Comparison with the state-of-the-art models

Many researches have been conducted on the topic of predicting students' performance. We have provided a detailed comparison between them and our proposed models in this section. Badr (2016) applied the AR mining technique to predict students' academic performance in two categories (Good, Pass). In this experiment, they used a real dataset consisting of 203 records and observed that the CARM algorithm had a higher accuracy rate of 67.33%. Sultana et al. (2018) used an online dataset consisting of 480 records with 16 characteristics gathered from Kaggle to measure students' academic performance. They applied various ML techniques, such as Multi-Layer Perceptron (MLP), LMT, RF, DT, K-star, IBK, Decision Table, NBTree, and Random Tree to predict grades in three categories (Low-Level, Middle-Level, High-Level), and observed that the MLP algorithm had a higher accuracy rate of 78.33%.Kumar et al. (2021) used an online dataset consisting of 480 records with 16 characteristics gathered from Kaggle. They applied various ML techniques such as Naive Bayes, MLP, Locally Weighted Learning (LWA), Decision Table, J48, and work enhancement, for these using ensemble methods such as AdaBoost, and Bagging. Their findings showed that the MLP with the AdaBoost algorithm achieved the highest accuracy rate at 80.83% in three categories (Low, Medium, and High). Hu and Rangwala (2019) used an online dataset consisting of 3,728 and collected from a CS course in Spring 2017 was used to predict the academic performance of students in two categories (High, and Low). The researchers in this work used ML and DL techniques and observed that the LSTM algorithm had a higher accuracy rate of 81.9%. Yakubu and Mohammed Abubakar (2022) aimed to reduce dropout levels by helping students predict performance in two categories and assess students' academic performance. They applied the LR to a 978 records dataset from (American University of Nigeria) AUN and observed that the LR algorithm had an accuracy rate of 83.5% in two categories (Pass, and Fail). Pujianto et al. (2020) aimed to reduce dropout rates by aiding students in predicting their course grades prior to enrollment. They used an online dataset gathered from Kaggle and applied KNN and DT and their findings indicated that the

Table 9 Comparison with previous research regarding prediction of student performance

Paper	Dataset Source	Dataset Size	ML& DL models	Highest algorithm accuracy score	Class label
Badr (2016)	Mathematics Department in the Faculty of Sciences, Real Data	203	Association Rule (Classification Association Rules Mining (CARM)) with support = 6 and confidence = 72	CARM algorithm achieved 67.33% accuracy	2 (Good, Bad)
Sultana et al. (2018)	Kaggle (Kalboard 360 dataset repository), Online Data	480	MLP, LMT, RF, DT, K-star, IBK, DT, NB Tree, and Random tree	MLP algorithm achieved 78.33% accuracy	3 (Low, Middle, High)
Kumar et al. (2021)	Kaggle (Kalboard 360 dataset repository), Online Data	480	NB, MLP, LWA, DT, J48, AdaBoost, and Bagging	MLP with AdaBoost achieved 80.83% accuracy	3 (Low, Medium, High)
Hu and Rangwala (2019)	Online Data for CS course	3,728	MLP and LSTM	LSTM achieved 81.9% accuracy	2 (High, Low)
Yakubu and Mohammed Abubakar (2022)	American University in Nigeria, Real Data	978	LR	LR achieved 83.5% accuracy	2 (Pass, Fail)
Pujianto et al. (2020)	Kaggle (Kalboard 360), Online Data	480	KNN and DT	DT algorithm achieved 71.09%	3 (Low, Middle, High)
The Proposed Model	the dataset was collected from FCIS, Real Data	3981	ML such as (DT, LR, SVC, LDA, QDA, KNN, RF, GB, XGBoost, LightGBM, Extra Trees) and DL such as (DANN, LSTM, and GRU)	SVC achieved a higher accuracy rate of 78.04%	3 (Low, Medium, High)

DT algorithm achieved a superior accuracy rate of 71.09% across three categories (Low, Middle, and High). As noted in Table 9, the authors conducted a comparison with previous research regarding predictions for student performance, aiming to predict grades in upcoming courses based on grades from previous courses in the early stages of the semester. They also developed recommendations for suitable courses and optimal department choices. Most previous research dealt with small online datasets ranging from 162 to 480 or 575 records, or small real datasets (Farissi et al., 2020; Badr, 2016; Sultana et al., 2018). In contrast, we utilized a large real dataset comprising 3981 records from FCIS. Additionally, the majority of prior studies focused either solely on ML or DL and applied either prediction or recommendation models. In this study, the authors incorporated two models. The first model involves prediction using both ML and DL techniques, while the second model is a recommendation system (Yağcı 2022), (M. Kumar et al. 2021. In this study, it was observed that the highest accuracy was achieved by SVC, with an accuracy of 78.04%. Based on these experimental results, our proposed model outperformed all other evaluated techniques.

Technical analysis of the study prediction algorithms

The ML models were evaluated using both a random hold-out method in Fig. 5 and 6-fold cross-validation in Fig. 7. Consistently across both methods, SVC emerged as the best-performing model, achieving high test accuracy (78.04% in hold-out and 76.16% in cross-validation) and F1-scores (75.37% and 73.35%, respectively). This indicates that SVC has strong predictive accuracy and generalization capability, making it suitable for real-world applications where stable performance is crucial.

Other algorithms, such as LR, RF, GB, and Extra Trees, also demonstrated solid performance, achieving high accuracy and F1-scores across both validation techniques. These models performed consistently well and could serve as effective alternatives to SVC if required, especially in ensemble configurations or when exploring further feature interactions.

The DT model consistently exhibited the lowest performance among the ML models, with test accuracies of 61.43% (hold-out) and 61.68% (cross-validation), and corresponding F1-scores around 60%. This suggests that DT struggles with capturing complex patterns in this dataset and may require significant tuning or ensemble methods to be viable in this context.

Among the DL algorithms, LSTM exhibited the highest test accuracy (76.16%) and F1-score (74.62%) in the hold-out evaluation as shown in Fig. 6, indicating that it may capture temporal dependencies well in the student performance data.

GRU performed closely to LSTM, suggesting it may be a viable alternative where computational efficiency is a priority, as it achieved a 74.40% accuracy and 72.65% F1-score.

DANN achieved lower scores (71.14% accuracy and 70.55% F1-score), suggesting that it may not be as effective for this dataset compared to the other DL methods.

The study recommendation model

Recommendations of courses and department based on the prediction model

As mentioned in the earlier sections, university students often struggle to choose the suitable department within their faculty and select the courses best suited for them, which will ultimately shape their career path after graduation. Providing a course recommendation model based on a course prediction model can solve this problem and benefit all stakeholders by enriching the academic performance of students, assisting faculty in their teaching and advising duties, and improving the university's operational efficiency and effectiveness. 9 presents the study recommendation model.

In this study, a prediction model has been developed to predict students' grades in subsequent years based on their actual grades from their first year. As shown in Fig. 9, the output of this prediction model is used as input for a recommendation model. This recommendation model suggests the most appropriate courses for the student, prioritizing those where the prediction model predicts the highest grades and considering the number of other courses available for enrollment if the student selects a particular course. After applying various machine learning and deep learning algorithms to predict student grades, the highest accuracy was achieved using the SVC algorithm, which was then utilized in the development of the recommendation model. The recommendation system takes the predicted grades for each course from the SVC-based prediction model and applies the Faculty of Computer and Information Sciences (FCIS) rules, such as the required GPA for enrollment in a course, the prerequisite courses needed, and the number of other available courses that can be opened by a specific course. The recommendation model then suggests the course with the highest predicted grade and the most additional courses available for registration. Additionally, the recommendation system recommends the student with the most suitable department to enroll in, based on the department with the highest number of courses where the student is expected to achieve high grades. Further details about the recommendation model are provided in the following section.

Description and example of the study recommendation model

Students often encounter difficulties when it comes to choosing the most suitable courses during their undergraduate studies. In academic life, the faculty offers

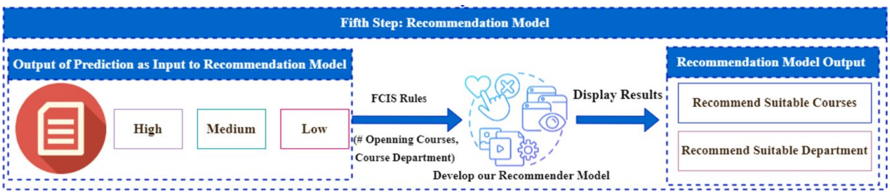

Fig. 9 The study recommendation model steps based on the developed prediction model and FCIS rules

students multiple courses to study. Students have various options to select from multiple courses, often guided by their future career knowledge and advice from teaching assistants, seniors, and other sources of guidance. Therefore, choosing the wrong courses can result in numerous problems and poor performance.

So, in this section, the authors present the second model in this work: a recommendation system to recommend suitable courses for students to solve the previous problems. As declared in the proposed work Fig. 4, the recommendation model utilizes the output of the first model, which is the prediction model, as input for the second model.

The study recommendation model offers two services for students. The first service is the recommendation of the most suitable course for him in terms of grade based on the study prediction model and the number of other courses that will be open by enrolling in this course based on FCIS course constraints. meaning that it recommends the highest course in terms of grade and the number of other courses that will be opened by it. As shown in Table 10, each course has a number of courses that can be opened and the student can enroll in it if he succeeds in the previous course. The second service is helping students choose which department to enroll in during their third academic year. FCIS at Mansoura University divides students' courses into three clusters. Each cluster represents a department (Information Systems (IS), Information Technology (IT), and

Table 10 The courses in the first and second levels with their departments and the number of opening courses for each course in the second level

Course Name	Level Number	# Opening Courses	Department Name
Integral and Derivative	Level 1	1	Information Systems (IS)
Information System	Level 1	1	
Web Design	Level 2	2	
Data Structures	Level 2	2	
Linear Algebra	Level 2	1	
Probability 2	Level 2	1	
Computer Graphics	Level 2	1	
Database	Level 2	1	
Physics	Level 1	1	Information Technology (IT)
English	Level 1	0	
Logic	Level 1	1	
Information Technology	Level 1	1	
Data Communication	Level 2	1	
Computer Science	Level 1	1	Computer Science (CS)
Discrete Mathematics	Level 1	1	
Fundamental of Programming	Level 1	2	
Probability 1	Level 1	1	
Object Oriented Programming	Level 1	2	
Operating System	Level 2	1	
Computer Architecture and Organization	Level 2	1	

Computer Science (CS), as shown in Table 10, which is an example of student courses and their departments in the first and the second academic years. Based on student course grades in the first academic year and the predicted course grades in the second academic year, the recommendation model will recommend the suitable department in which the student has the highest grades in courses. As shown in Fig. 10 the student has the highest grades (high) in five courses (Web Design, Probability 2, Computer Graphics, Database, Computer Architecture, and Organization). The first four belong to the information systems department, and the last belongs to the computer science department. Also, the model has predicted medium grades in four courses (Data Structures, Linear Algebra, Operating Systems, and Data Communication) for the test student. The first two belong to the information systems department, the third to the computer science department, and the last to the information technology department. There are no courses with low grades for this test student. So, the recommendation system recommends the information systems department because the highest grades were in this department, as shown in Fig. 10.

Figure 10 is an example of the study recommendation model. We took a sample of degrees for a student in the first level. Then the prediction model predicts the degrees of the next level in three levels (high, medium, and low), as we declared previously. Finally, depending on the actual grades in the first academic year, the previously predicted grades, and the constraints of FCIS (the FCIS rule), the recommendation system recommends the suitable department for the next levels.

Algorithm 1 shows the steps of the training classification model for the prediction model and the steps the study recommendation model.

```
-------------------Level One Degrees as an Input For Our Prediction Model.--------------------------
Physics =  97  & CS =  75  & D-Math =  71  & English =  68
Fund_prog =  78  & Logic =  87  & Integral =  87  & IS =  87
Prob1 =  77  & IT =  62  & OOP =  70
----------------------------------------------------------------------------------------------------
Our Recommendation System Recommends three levels of courses depending on the FCIS rules and the predicted degrees based on the
study preiction model.
Good Luck for your choose.!
----------------------------------------------------------------------------------------------------

Level One is (High), and this level Contain Courses Degrees from 80 to 100
----------------------------------------------------------------------------------------------------
The Predicted Grade for  Web Design  Course Is  ['High'] & Open  2  Courses To Next Levels!.
The Predicted Grade for  Probability 2  Course Is  ['High'] & Open  1  Courses To Next Levels!.
The Predicted Grade for  Computer Architecture and Organization  Course Is  ['High'] & Open  1  Courses To Next Levels!.
The Predicted Grade for  Computer Graphics  Course Is  ['High'] & Open  1  Courses To Next Levels!.
The Predicted Grade for  Database  Course Is  ['High'] & Open  1  Courses To Next Levels!.
----------------------------------------------------------------------------------------------------

Level Two is (Medium), and this level Contain Courses Degrees from 50 to 79
----------------------------------------------------------------------------------------------------
The Predicted Grade for  Data Structures  Course Is  ['Medium'] & Open  2  Courses To Next Levels!.
The Predicted Grade for  Operating System  Course Is  ['Medium'] & Open  1  Courses To Next Levels!.
The Predicted Grade for  Linear Algebra  Course Is  ['Medium'] & Open  1  Courses To Next Levels!.
The Predicted Grade for  Data Communication  Course Is  ['Medium'] & Open  1  Courses To Next Levels!.
----------------------------------------------------------------------------------------------------

Level Three is (Low), and this level Contain Courses Degrees from 0 to 49
----------------------------------------------------------------------------------------------------

----------------------------------------------------------------------------------------------------
---------------------------------------Department Selection-----------------------------------------
The Predicted Department For You is The Information Systems Department
----------------------------------------------------------------------------------------------------
```

Fig. 10 Example of the study recommendation model. The recommendation model recommends a suitable department based on the study prediction model and FCIS rules

Algorithm 1 The training classification prediction and recommendation models.

1. **Input:** Students grades X of (MU-dataset) where X € [0:100]
2. **Output:** A recommendation model utilizes the output of the first model, which is the prediction model, as input for the second model.
3. **Steps:**
4. **Data pre-processing steps:**
5. Data cleaning:
6. **Foreach** record **do:**
7. Detect outliers.
8. Remove noise.
9. Complete missing values.
10. **End**
11. Data discretization:
12. Convert Y values (numerical) into three ordinal values (high, medium, low).
13. Feature encoding
14. Data normalization
15. **Foreach** record **do:**
16. Determine the new value of Xscaled:
17. $$X(\text{ Value Scaled }) = \frac{x-\mu}{\sigma}$$
18. **End** Foreach.
19. **End** pre-processing **steps**.
20. Applying two different methods of model validation.
21. Random Hold-Out.
22. Stratified 6-Fold Cross-Validation.
23. Building the study prediction model.
24. Employ different ML and DL models M where M € [DT, LR, SVC, LDA, QDA, KNN, RF, GB, XGBoost, LightGBM, Extra Trees, DANN, LSTM and GRU].
25. **Foreach** M **do:**
26. Assign the appropriate parameters for M.
27. **Foreach** FCIS course C **do:**
28. Predict the students' grades Y where Y € [High, Medium, Low].
29. Calculate the accuracy, precision, recall, and F1-Score for each M
30. **End** Foreach.
31. Compare the performance for each M
32. **End** Foreach.
33. Select the best performance model as the base model for prediction and recommendation (SVC).
34. Building the study recommendation model.
35. Assign the SVC model M as the prediction model for students' grades Y.
36. Assign the actual students' grades in the first academic year to M.
37. **Foreach** FCIS course C **do:**
38. Assign the number of other opening courses by C, C department.
39. predicate the Y value using M.
40. Assign FCIS rules r where r € [required GPA, the prerequisite courses needed.]
41. **End** Foreach.
42. **Foreach** FCIS course C in the following year **do:**
43. Compare the predicted grades.
44. Compare the number of other courses that can be opened by C.
45. **End** Foreach.
46. **IF** the student has the required FCIS rules to enroll in course **do:**
47. Recommend the course with the highest predicted grade and the most additional courses available for registration.
48. Recommend the department with the highest number of courses where the student is expected to achieve high grades.
49. **End** IF.
50. **End.**

Conclusion and future work

Student achievement and performance are important characteristics that have been closely monitored by the education sector, notably by institutional administration. Institutions normally consider and regularly emphasize the necessity of identifying students' performance. Various courses are available (openly) for selection at institutions, including mandatory and elective courses. Students are typically required to enroll in mandatory courses, while they have the flexibility to select elective courses from a range of optional course groupings available in the curriculum. Students face challenges in selecting appropriate courses and departments to enroll in, which is recognized as the most essential aspect of preventing career failure because enrolling in inappropriate courses may result in failure in future work. Developing a recommendation model that recommends students with the most appropriate courses and departments based on FCIS rules and the predicted grade outputs for upcoming courses based on their grades in preceding courses from the first academic year by another predictive model is the primary objective of this study.

The EDM has shown great performance in analyzing educational data and extracting hidden knowledge, which leads to great assistance in improving the learning process. The study applied different EDM methods for building a predictive model that predicts students' grades in upcoming courses based on their grades in previous courses. In this study, we applied eleven ML algorithms and three DL algorithms, namely DT, LR, RF, SVC, LDA, QDA, KNN, GB, XGBoost, Light GBM, Extra Trees, DANN, LSTM, and GRU, using a real dataset (MU-dataset) collected from the FCIS at Mansoura University. The current study's dataset includes 3981 records with 11 attributes indicating all students' grades in all courses they took during their first and second academic years. These factors pertain to the academic realm and are specifically connected to students' grades in their first-year courses. Various data pre-processing methods were implemented on the dataset for this study. These methods encompassed data cleaning, data discretization, feature encoding, and data normalization.

In addition, this study applied both K-fold cross-validation and random hold-out techniques for model validation. The findings (results) indicate that, for both the ML and DL algorithms, the random hold-out method outperforms the k-fold cross-validation method. SVC exhibited the most outstanding performance in predicting students' academic performance, achieving a multi-classification accuracy of 78.04%. Two models were developed for university students. The first is a model that predicts students' academic grades in upcoming courses. The second is a recommendation model that recommends the highest-graded courses and the number of other courses that will be open to students by enrolling in this course. In addition, it recommends a suitable department in which students will have the highest grades in their courses. The obtained results highlight the importance of course recommendation model, the developed course recommendation model based on the developed prediction model offers an important service for all stakeholders, enhancing the academic experience for students, and enabling the faculty and university to operate more efficiently and effectively. For students, it improves their academic performance by suggesting courses where the student is more likely to succeed based on past performance and learning style, helping in identifying prerequisite courses that can strengthen a student's understanding of advanced topics, and minimizing the chances of taking unnecessary or redundant courses, saving both time and cost. For faculty, it enables faculty to guide students toward courses that align with their academic performance. For universities, it improves graduation rates by guiding students toward courses that match their abilities and can get higher grades in them, contributing to higher graduation rates. Collectively, these services contribute to a more effective and responsive educational environment that provides a competitive advantage in attracting and retaining students.

The study provides a great tool for helping students select their career paths based on their academic features and highlights the importance of EDM in decision-making. Furthermore, the prediction model results highlight the significance and underscore the effectiveness of DL and ML algorithms in dealing with educational data and predicting students' academic performance. also, indicates the effective performance of random hold-out techniques for model validation.

SVC's high performance across both hold-out and cross-validation indicates its robustness and suitability for the prediction task, making it a logical choice for the recommendation model. SVC can also be combined with ensemble approaches such as RF, GB, Extra Trees, and LightGBM to capture diverse patterns within the data, potentially enhancing the robustness and accuracy of predictions. On the other hand, the DL models, while competitive, generally did not outperform the top ML models significantly. However, GRU's relatively high accuracy and balanced precision-recall trade-off suggest it might be useful in applications where sequence modeling is necessary or where the dataset is larger and more complex.

Given SVC's strong and consistent results, its selection for the recommendation system is justified. The DL models, particularly GRU, could be revisited for future versions of the recommendation system, especially if enhanced temporal modeling becomes essential.

In the future, the authors intend to enhance the model's performance by augmenting the dataset with additional academic and non-academic data. More ensemble techniques will be applied to the proposed model. In addition, the authors will make a mobile application for FCIS students that allows them to enroll in courses, tells them exam and lecture dates, and saves other academic data for managing and improving the learning process in FCIS.

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Declarations

Conflict of interest The authors declare no conflict of interest.

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